

## **FINANCIAL MANAGEMENT II (ACC 423)**

### **LECTURE NOTE**

#### **INTRODUCTION**

Emerging capital markets all over the world have a history and Nigeria is not in any way different. The Nigerian capital market represents a small but growing proportion of the nation's economy. The development of capital market institution in an emerging economy like Nigeria seeks to look not only on how it came to be established but also how this institution has evolved to impact upon the Nigerian economy. The Nigerian capital market is principally a market for long-term investments where corporate equities and long-term debt securities are issued and traded. It is a market that is regulated by the Securities and Exchange Commission (SEC), which is the apex regulatory body of the Nigerian Capital Market. The capital market is so vital to the development of any economy that no country can risk its existence without putting in place measures to regulate the activities of the market.

Consequently, government, at various points in time sought to promote an orderly growth and development of the market for viable investment opportunities. It is against this background that the need to consider the establishment and operation of the Security and Exchange Commission becomes imperative.

History of capital market in Nigeria can be traced back to 1946, with the floatation of 300,000 pounds bond by the then colonial government to implement its 10-year development plan. In 1959, the Lagos Stock Exchange was established as a private company limited by guarantee, and commenced operation in 1961.

#### **CAPITAL MARKET:**

As stated earlier, The Nigerian capital market is principally a market for long-term investments where corporate equities and long-term debt securities are issued and traded. Capital market consists of two (2) types i.e primary and secondary.

- a. Primary Market is the market for new shares or securities while,
- b. Secondary Market deals with the exchange of prevailing or previous-issued securities among investors.

Examples of highly organized capital markets are the New York Stock Exchange, American Stock Exchange and London Stock Exchange.

#### **The Security and Exchange Commission (SEC)**

The Security and Exchange Commission (SEC) is the main regulatory institution of the Nigerian capital market. It is supervised by the Federal Ministry of Finance.

The Security and Exchange Commission (SEC) has number of roles including:

1. Registers all securities to be offered for sale to, or for subscription by the public.
2. Approves timing of new issue.
3. Maintains surveillance over dealings in securities.
4. Registers all stock exchange including their branches and capital market operators, and
5. Protects the security market against any manipulations including insider trading.

Companies wishing to be admitted to the official list of the securities and exchange must, in addition to complying with the exchange's listing rules comply with relevant provisions of the Company and Allied Matters Act 1990 and the Investment Securities Act, 2007 Rules and Regulations.

### **The Nigerian Stock Exchange (NSE):**

The Nigerian Stock Exchange (NSE) provides exchange services. The company offers listing and trading services, as well as electronic clearing, settlement and delivery, and market data dissemination services. The Nigerian Stock Exchange (NSE) serves customers throughout Nigeria. It operates Automated Trading System (ATS) platform with a central order book that allows trading license holders to participate on equal terms, competing on hierarchical basis of price cross and time priority.

The Nigerian Stock Exchange (NSE) has number of roles including:

1. Provides the platform for buying and selling of existing securities
2. Provides liquidity for investors
3. Regulates the activities of the stock brokers
4. Encourage transactions in the new issue market
5. Helps to spread promoters' risks

### **Issuing house:**

Issuing house is simply a financial house formed for the purpose of serving as a financial adviser to companies, and these companies may be either quoted companies and/or unquoted companies. Issuing house can also serve as financial adviser to federal government agencies and government institutions and as well, to state governments in Nigeria, that seek to raise additional money (fund) for the purpose of developing (to finance) their capital projects, for example, capital projects such as, roads projects, and schools projects, and hospitals project, and other infrastructure projects among other projects from the capital market. Issuing are also saddled with the responsibility of issuing out bonds to seekers. Though there are requirements to be met before a bond can be issued out.

Issuing houses act as financial intermediaries in:

- i. Providing financial advisory services
- ii. Issuing of bonds
- iii. Stockbroking
- iv. Underwriting of securities
- v. Pricing of new issues (along with the issuing company)
- vi. Bringing new issues to the market
- vii. Determining the timing of new issues (subject to the approval of SEC)
- viii. Raising of working capital to companies
- ix. Working with regulatory bodies and authorities
- x. Provision of tailor-made and custom solutions to clients

### **Participants in the capital markets:**

The participants in the capital markets are those who are actively involved in the purchase and sale of stocks and shares and those institutions that are involved as regulatory authorities. They include in the main:

- a. Individuals and companies
- b. Banks and other financial institutions
- c. Issuing houses
- d. Stockbroker
- e. The Nigerian Stock Exchange (NSE)
- f. Securities and Exchange Commission (SEC)
- g. Central Bank of Nigeria (CBN)
- h. Registrars
- i. Fund/ portfolio managers
- j. Receiving Bankers
- k. Trustees
- l. Capital market consultants
- m. Central Securities Clearing System

### **Methods of listing Securities:**

The methods of listing securities on the NSE include among others:

**Public issue:** This is also known as ‘offer for subscription’ an invitation by or on behalf of a company or other authority to the public, for subscription of securities at a fixed price.

**Offer for Sale:** An offer to the public by or on behalf of a shareholder, the proceeds which will go to vendor(s).

**Placing:** Sale by a broker to his clients, of securities, which have previously been purchased or subscribed for.

**Right offer/issues:** A privilege offer/issue to existing shareholders to acquire proportionately additional shares in the company usually at a special price;

**Private placement:** A per- selection placement to few groups of people e.g family and friends or investors and institutions especially in risky ventures.

**Capitalization issue:** a bonus/ scrip issue to existing shareholders.

**Tender:** An offer of specific quantity of shares and stock to the public by or on behalf of a company or other authority or a third party for bidding;

**Introduction:** The listing of securities already widely held;

**Conversion:** An exchange for or conversion of securities into other classes of securities;

**Option:** An offer to buy or sell some shares at an agreed price and time;

## Public issues of Bonds and Debentures:

### Bonds market:

Bond is an interest-bearing debt security and instrument, that is issued by corporate organizations, as well as government for development of infrastructures. Most bonds are issued for a fixed period of time (maturity) after which they are redeemed by the issuer, usually at their face value.

### Debenture stock:

A debenture is a type of debt instrument that is not backed by any collateral and usually has a term greater than 10 years. Debentures are backed only by the credit worthiness and reputation the issuer. Both corporations and governments frequently issue debentures to raise capital or funds.

Also, debenture is a long-term security yielding a fixed rate of interest, unsecured by collateral often used by private companies to raise capital.

### Cost of New Equity

This is also known as floatation cost is the cost associate with issuing new stock to the capital market in order to raise more funds.

Floatation cost will increase the cost of new equity, as a result, it will be higher than the existing equity in the same company. It includes others costs such as underwriting fees, registration, legal fees, and other costs. these fees will impact the amount of capital to be issued.

These costs mostly the percentage of share price, so we will calculate by applying the percentage below formular.

$$\text{Cost of new equity} = \frac{D_1}{P_0 (1 - f)} + g$$

Where:

$D_1$  = dividend at the end of next year,       $P_0$  = the issued share price

$f$  = the ratio of the floatation cost,       $g$  = dividend growth rate

### Example:

Bams Ltd. decided to issue N500 millions of new common stocks to the market, they are issued at N100 per share and the broker charge fee 5% over the share price. Based on historical data, the annual dividend expected to be N5 per share and it will grow at 3% rate. Please calculate the cost of new equity.

### Lease vs Buy decision:

A company might be faced with a decision about whether or not to acquire a new non-current asset, and if the asset is acquired, whether the acquisition should be financed with a bank loan or by means of a lease arrangement. This is usually referred to as 'lease versus buy' decision.

The decision should be considered in two (2) separate stages.

1. The acquisition decision
2. The finance decision

## **Types of lease:**

IAS 17 described two types of lease (with each type being accounted for in a different way):

- a. Finance lease and
- b. Operating lease

A lease is classified as a finance lease if it transfers substantially all the risks and rewards incidental to ownership. A lease is classified as an operating lease if it does not transfer substantially all the risks and rewards incidental to ownership.

Whether a lease is a finance lease or an operating lease depends on the substance of the transaction rather than the form of the contract.

The legal form of a finance lease is that the lessor is the legal owner of the leased assets.

The economic substance of a finance lease is that the lessee has all the benefits and costs associated with ownership of the asset. The finance lessee is in the same position as it would have been if it had borrowed money to buy the asset itself. That is why such leases are called finance leases, they provide finance for the use of an asset.

## **IFRS 16**

IAS 17 has been superseded by IFRS 16. Under the new rules in IFRS 16 lessees must capitalise all leases in their financial statements. The distinction between operating and finance leases no longer applies for a lessee.

However, the distinction between operating and finance leases continues to apply from the lessor's view as do the finance lease indicators given above.

This all might sound to be irrelevant for this paper but lease or buy decisions are based on relevant cash flows. Taxation is a relevant cash flow and the tax legislation continues to make reference to operating and finance leases in the rules on tax deductibility. So in this respect the distinction is still important for lessees because it affects the deductibility of their lease payments.

## **Tax savings on capital allowances**

Nigerian rules in regard to capital allowances are as follows:

A company that buys an asset to use in its business can claim capital allowances.

A company that leases an asset under a finance lease can also claim capital allowances.

A company leasing an asset under an operating lease cannot claim capital allowances.

These are claimed by the lessor.

Therefore, the tax effect of the capital allowances is incremental to the financing decision between borrowing and buying an asset and leasing the asset under an operating lease.

The tax effect of capital allowances is not incremental to the financing decision between borrowing and buying an asset and leasing the asset under a finance lease.

## **Example: Lease or buy decision**

HND Limited .....

### **Advantages of leasing**

1. Balanced cash outflow
2. Quality Assets
3. Better usage of Capital
4. Tax benefit
5. Off-balance sheet Debt
6. Better planning
7. No risk of Obsolescence
8. Termination right
9. Low capital expenditure

### **Disadvantages of leasing**

1. Lease Expenses
2. No ownership
3. Limited financial benefits
4. Reduced return for equity holders
5. Debt
6. Limited access to other loans
7. Processing and documentation
8. Maintenance of the asset
9. Limited tax benefit

### **The efficiency of capital markets and fair prices**

Investors in securities such as shares and convertible bonds want to be confident that the price they pay for their securities is a fair price. In order for market prices to be fair, it is important that the stock market should be able to process the relevant available information about companies and that investors should have immediate access to this information and act on it when making decisions about buying and selling shares.

The efficient markets hypothesis provides a rational explanation of how share prices change in organised stock markets. The hypothesis is based on the assumption that share prices change in a logical and consistent way, in response to new information that becomes available to investors. The **speed** with which share prices change depends on how quickly new information reaches investors, and this varies with the **efficiency** of the market.

### **The nature of capital market efficiency**

There are 4 types of capital market efficiency

**Operational efficiency** - A capital market is efficient operationally when transaction costs for buying and selling shares are low, and do not discourage investors from taking decisions to buy or sell.

**Informational efficiency** - A capital market is efficient 'informationally' when available information about companies is processed and made available to investors.

**Pricing efficiency** - A market has pricing efficiency when investors react quickly to new information that is made available in the market, so that current share prices are a fair reflection of all this information. For pricing efficiency to exist, a capital market must also be operationally and informationally efficient.

**Allocational efficiency** - When there is allocational efficiency in a capital market, available investment funds are allocated to their most productive use. Allocational efficiency arises from pricing efficiency.

Research into stock market efficiency focuses on pricing efficiency.

**Efficiency** therefore refers to the speed with which information is made available to the market, and the response of market prices to this information. In an efficient market, all investors are reasonably well informed at the same time about new developments that might affect market prices, so that some investors with ‘insider knowledge’ cannot exploit their knowledge to make profits at the expense of other investors.

If all relevant information is made available to all investors at the same time, all investors are able to make decisions at the same time about buying or selling investment, and about whether current prices are too high or too low.

Although the concept of market efficiency applies to all financial markets, it is probably most easily understood in the context of equity shares and the equity markets.

### **The purpose of the efficient market hypothesis (EMH)**

The efficient market hypothesis (EMH) is a theory of market efficiency, based on research into share price behaviour in stock markets. The purpose of this research is to establish the extent to which capital markets show pricing efficiency.

According to this theory there are three possible levels or ‘forms’ of market efficiency:

- Weak form efficiency
- Semi-strong form efficiency,
- Strong form efficiency.

Each financial market can be categorised as being weak form, semi-strong form or strong form efficient.

In equities markets, the way in which share prices move in response to available information varies according to the efficiency of the market.

### **Weak form efficiency**

The efficient markets hypothesis states that when a market has weak form efficiency, share prices respond to the publication of historical information, such as the previous year’s financial statements.

When the market displays a weak form, it also means that the current share price embodies all the historical information that is known about the company and its shares, including information about share price movements in the past. Until the next publication of more historical information about the company, there is no other information about the company that will affect the share price in any obvious way.

The weak form suggests that the current price reflects all past prices and that past prices and upward or downward trends in the share price cannot be used to predict whether the price will go up or down in the future. Share prices do rise and fall, with supply and demand in the market, but the next price movement is equally likely to be up as down.

### **Random walk theory (versus Chartism)**

A weak form of stock market efficiency is consistent with the **random walk theory**. This is the theory that share prices move up and down randomly over time, in response to the arrival of favourable or unfavourable information on the market.

Random walk theory is opposed to the view that future share price movements can be predicted from patterns of share price movements in the past, since patterns repeat themselves, and historical trends can be used to predict future trends. Some stock market analysts believe

that they can predict future movements in share prices from recognisable patterns of share price movement. These analysts are sometimes called **chartists**, because recognisable patterns of share price movements can be illustrated by graphs or charts of share prices over a period of time. Chartism does not have a rational justification.

### **Semi-strong form efficiency**

When a market has semi-strong form efficiency, current share prices reflect all publicly-available information about the company and its prospects, in addition to historical information. For example, share prices might respond to a new announcement by a company about its trading prospects for the remainder of the year. Similarly, the share price might also respond to an announcement that the company is seeking to make a new acquisition, or a major new investment.

If a market displays semi-strong form efficiency, share prices should move when new information becomes available to the public, but not before. For example, if a company is planning a major acquisition, the share price should not be affected by unconfirmed rumours in the market. However, the share price will react to the official announcement of a takeover bid by a company.

It also means that individuals who have access to information that has not yet been made public (**‘inside information’**) will be able to buy or sell the shares in advance of the information becoming public, and make a large personal profit. This is because the inside information will indicate whether the share price is likely to go up or down, and the individual can buy or sell accordingly.

Using inside information to make a personal profit from trading in shares is called insider dealing, which is illegal in countries with well-established stock markets.

### **Strong form efficiency**

When a market has strong form efficiency, current share prices reflect **all** relevant information about the company as soon as it comes into existence, even if it has not been made publicly- available. In other words, the share price reflects all inside information as well as publicly-available information. The market is so efficient that all information is immediately transmitted throughout the market instantly, and all investors have access to this same information.

If the stock market has strong form efficiency, it is impossible for individuals to profit from insider trading, because there is no inside knowledge that the market has not already found out about.

In practice, research suggests that most markets have weak form efficiency, but some well- developed markets such as the New York Stock Exchange and London Stock Exchange are semi-strong form efficient.

### **Example:**

#### **Efficient markets hypothesis**

A company decides to undertake a major capital investment. The investment will be in a five-year project, and over the course of the five years, the company’s directors believe that the net profits will add ₦125 million to the value of the company’s shares.



The company made the decision to invest on 1<sup>st</sup> October Year 1, and the first year of profits from the investment will be Year 2. It announces the investment and the expected benefits to the stock market on 1<sup>st</sup> December. It is assumed that the stock market investors believe the company's estimate that the project will add ₦125 million to share values.

### **Strong form efficiency**

If the stock market has strong form efficiency, the company's share price should go up on 1<sup>st</sup> October, as soon as the decision to invest is made. The total increase in share value should be ₦125million.

### **Semi-strong form efficiency**

If the stock market has semi-strong form efficiency, the share price should go up on 1<sup>st</sup> December, when the investment and its expected benefits are announced to the market and so become public information. (Between 1<sup>st</sup> October and 1<sup>st</sup> December, the information is 'inside information').

### **Weak form efficiency**

If the stock market displays weak form efficiency, the share price will not be affected by the announcement on 1<sup>st</sup> December Year 1. The share price will eventually respond, after each of the next five years, when the actual historical profits of the company, including the profits from the new investment, are announced.

### **Implications of strong capital market efficiency**

There are several theoretical implications of market efficiency. If a capital market has strong efficiency:

- ☐ Share prices will be fair at all times and reflect all information about a company. This means that there is no 'good time' or 'bad time' to try issuing new shares or bonds.
- ☐ Companies will gain no benefit from trying to manipulate their financial results and present their performance and financial position in a favourable light. In a market with strong-form efficiency, investors will see through the pretence and will understand the true financial position of the company.
- ☐ For investors there will never be any 'bargains' in the stock market, where share prices are under-valued. Similarly there will be no over-priced shares that clever investors will sell before a share price fall.
- ☐ If the capital market has strong form efficiency, if a company invests in any new capital project with a positive net present value, the share price should respond by going up to reflect the increase in the value of the company represented by the project NPV.

### **Portfolio analysis: Risk and Investment**

#### **Risk and return in investments**

Investors invest in shares and bonds in the expectation of making a return. The return that they want from any investment could be described as:

- A return as reward for providing funds and keeping those funds invested, plus
- A return to compensate the investor for the risk.

As a basic rule, an investor will expect a higher return when the investment risk is higher.

### **What is investment risk?**

Investors in bonds, investors in shares and companies all face investment risk. In the case of **bonds**, the risks for the investor are as follows:

- The bond issuer may default, and fail to pay the interest on the bonds, or fail to repay the principal at maturity.
- There may be a change in market rates of interest, including interest yields on bonds. A change in yields will alter the market value of the bonds. If interest rates rise, the market value of bonds will fall, and the bond investor will suffer a loss in the value of his investment.

In the examination, you might be told to assume that debt capital is risk-free for the purpose of analysing the cost of equity. In practice however, only government debt denominated in the domestic currency of the government is risk-free.

In the case of **equity shares**, the risks for the investor are that:

- The company might go into liquidation; or
- Much more significantly, the company's profits might fluctuate, and dividends might also rise or fall from one year to the next.

For investors in equities, the biggest investment risk comes from uncertainty and change from one year to the next in annual profits and dividends. Changes in expected profits and dividends will affect the value of the shares. Bigger risk is associated with greater variability in annual earnings and dividends.

**When a company invests** in a new project, there will be an investment risk. This is the risk that actual returns from the investment will not be the same as the expected returns but could be higher or lower. This investment risk for companies is similar to the investment risk facing equity investors.

Some types of investment are more risky than others because of the nature of the industry and markets. For example, investments by a supermarkets group in Strategic financial management building a new supermarket is likely to be less risky than investment by an IT company in a new type of software. This is because the IT business is inherently more risky than the supermarkets business. When business risk is higher, returns are less predictable or more volatile, and the **expected returns should be higher to compensate for the higher business risk**.

### **Measuring risk as a variance or standard deviation of expected returns**

The risk of variations in annual profits and dividends can be measured. When annual returns can differ by a large amount from normal or expected (average) returns, and there are considerable differences in returns from one year to the next, returns are **volatile**.

High volatility in returns is associated with high investment risk.

Risk can be measured statistically, either from an analysis of historical returns achieved in the past, or from probability estimates of returns in the future. The measurement of risk is a measurement of the volatility of possible returns.

This volatility can be measured as either the variance or standard deviation of expected returns.

For an investment, we can measure both:

- an expected or average annual return; and
- a variance or standard deviation of the returns. (The standard deviation is the square root of the variance).

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**Formula: Variance**

$$\sum p(r - r)^2$$

**Where:**

p = the probability of a given return on the investment r

= the amount of that return

r = the average expected return on the investment. This is the expected value (EV) of the return.

**Example: Measuring risk as a variance or standard deviation of expected returns**

Based on historical analysis, it has been established that the probable returns on equity investment A are:

| Probability p | Return r |
|---------------|----------|
|               | %        |
| 0.2           | 2        |
| 0.5           | 6        |
| 0.3           | 8        |

**Example: Measuring risk as a variance or standard deviation of expected returns**

BAMS Ltd. has a two security portfolio namely security A and security B with each having the following returns and probability:

| Probability | Security A<br>Returns | Security B<br>Returns |
|-------------|-----------------------|-----------------------|
| 0.1         | 15%                   | 10%                   |
| 0.8         | 25%                   | 30%                   |
| 0.1         | 35%                   | 50%                   |

**Required:** Determine the:

- Expected returns from each security.
- Standard deviation of the returns from each security.
- The security with the highest risk.

### Concept of return

Investors hold securities (e.g. shares, bonds, etc.) because they hope for positive returns. Purchasers of ordinary shares, for examples are attracted by two components of return, first, the anticipated dividend(s) payable during the holding period, and second, the expected capital gain.

In general, for any holding period, the percentage return, R, from holding a stock is:

$$R = \frac{(EV - BV) + D}{BV} \times 100$$

Where:

EV = Value of the stock at the end of the holding period

BV = Value of the stock at the beginning of the holding period

D = Dividend received during the holding period

### Example: Calculating return

You bought a stock for ₦150 and it is now worth ₦55. During the period, you received a dividend at ₦2.

Calculate:

- i) the total return
- ii) the capital gain yield
- iii) the dividend yield

### Expected return from a two – asset portfolio

The expected return of a portfolio consisting of assets A and B is the weighted average of the returns from the two assets.

### Formula: Expected return from a two – asset portfolio

$$R_P = W_A R_A = W_B R_B$$

Where:

$W_A$  = % of funds, or weight, committed to asset A

$W_B$  = % of funds, or weight, committed to asset B

$R_A$  = the expected return from asset A

$R_B$  = the expected return from asset

### Example: Expected return from a two-asset portfolio

You have N1,000,000 to invest in stock A & B.  
The following is given for the two stocks.

| Stock | Current Price | Expected in price in 1 yr. | Expected dividend |
|-------|---------------|----------------------------|-------------------|
| A     | ₦40           | ₦48                        | ₦0                |
| B     | ₦50           | ₦63                        | ₦2                |

- You have decided to invest ₦750,000 on stock A and ₦250,000 on stock B. Calculate the expected return on your portfolio.
- Now assume you desire an expected return of 28% on your portfolio, how would you invest you ₦1,000,000?
- Alternatively, assume you desire a return of 36%, how much should you invest in each stock.

### Diversification to reduce risk

To a certain extent, an investor can reduce the investment risk – in other words, reduce the volatility of expected returns – by diversifying his investments, and holding a portfolio of different investments.

Creating a portfolio of different investments can reduce the variation of returns from the total portfolio, because if some investments provide a lower-than- expected return, others will provide a higher-than-expected return. Extremely high or low returns from the overall portfolio are therefore less likely to occur.

Similarly, a company could reduce the investment risk in its business by diversifying, and building a portfolio of different investments. However, it can be argued that there is no reason for a company to diversify its investments, because an investor can achieve all the diversification he requires by selecting a diversified portfolio of equity investments.

An investment portfolio consisting of all stock market securities (excluding risk-free securities), weighted according to the total market value of each security, is called the **market portfolio**.

The degree of reduction in risk due to diversification depends on how the returns on the investments change together in response to changing economic conditions. For example, if the returns on one investment fall as the returns on another rise they could be combined to reduce volatility (risk).

There are two important statistical measures used to describe the association between two variables. They are:

- covariance;
- correlation coefficient.

**Portfolio theory:**

Portfolio theory is concerned with how investors should build a portfolio of investments that gives them a suitable balance between return and investment risk. Portfolio theory provides a theoretical basis for the capital asset pricing model, which is an important model in financial management.

Risk averse investors will accept risk provided that they are compensated for it by an adequate return. However if they could reduce their risk without a reduction in return this would be even better.

This is possible by the process of diversification.

**Systematic and unsystematic risk**

Although investors can reduce their investment risk by diversifying, not all risk can be eliminated. There will always be some investment risk that cannot be eliminated by diversification.

- i. When the economy is weak and in recession, returns from the market portfolio as a whole are likely to fall. Diversification will not protect investors against falling returns from the market as a whole
- ii. Similarly, when the economy is strong, returns from the market as a whole are likely to rise. Investors in all or most shares in the market will benefit from the general increase in returns.

Therefore there are two types risk:

**Unsystematic risk**, which is risk that is unique to individual investments or securities, that can be eliminated through diversification

**Systematic risk or market risk:** This is risk that cannot be diversified away, because it is risk that affects the market as a whole, and all investments in the market in the same way.

**Implications of systematic and unsystematic risk for portfolio investment**

The distinction between systematic risk and unsystematic risk has important implications for investment.

- Investors expect a return on their investment that is higher than the risk-free rate of return (unless they invest 100% in risk-free investments).
- The higher expected return is to compensate investors for the higher investment risk.
- By diversifying, and investing in a wide range of different securities, investors can eliminate unsystematic risk. This is because if some investments in the portfolio perform much worse than expected, others will perform much better. The good-performing and poor-performing investments 'cancel each other out'.
- In a well-diversified portfolio, the unsystematic risk is therefore zero. Investors should therefore not require any additional return to compensate them for unsystematic risk.
- The only risk for which investors should want a higher return is systematic risk. This is the risk that the market as a whole will perform worse or better than expected.

## **CAPITAL ASSET PRICING MODEL (CAPM)**

### **Systematic risk in securities**

The capital asset pricing model (CAPM) establishes a relationship between investment risk and expected return from individual securities. It can also be used to establish a relationship between investment risk and the expected return from specific capital investment projects by companies.

The CAPM was explained in an earlier chapter as a model for estimating the cost of equity in a company, i.e. the returns that are required by equity investors. This section explains some of the assumptions and components of the CAPM.

As explained in the previous section, systematic risk is risk that cannot be eliminated by diversifying. Every individual security, with the exception of risk-free securities, has some systematic risk. This is the same systematic risk that applies to the market portfolio as a whole, but the amount of systematic risk for the shares in an individual company might be higher or lower than the systematic risk for the market portfolio as a whole.

Since investors can eliminate unsystematic risk through diversification and holding a portfolio of shares, their only concern should be with the systematic risk of the securities they hold in their portfolio. The return that they expect to receive should be based on their assessment of systematic risk, rather than total risk (systematic + unsystematic risk) in the security.

The CAPM assumes that investors hold diversified investment portfolios and are therefore concerned with systematic risk only and not unsystematic risk.

The systematic risk of a security can be compared with the systematic risk in the market portfolio as a whole.

- A security might have a higher systematic risk than the market portfolio. This means that when the average market return rises, due perhaps to growth in the economy, the return from the security should rise by an even larger amount. Similarly, if the average market return falls due to deterioration in business conditions, the return from the security will fall by an even larger amount.
- A security might have a lower systematic risk than the market portfolio, so that when the average market return rises, the return from the security will rise, but by a smaller amount. Similarly, when the average market return falls, the return from the security will also fall, but by a smaller amount.

A risk-free security has no systematic risk, because returns on these securities are unaffected by changes in market conditions. The shares of every individual company, however, have some systematic risk.

### **The beta factor of a security**

The systematic risk for an individual security is measured as a beta factor. This is a measurement of the systematic risk of the security, in relation to the systematic risk of the market portfolio as a whole.

The beta factor for the market portfolio itself = 1.0

## Formula for the CAPM

The formula for the capital asset pricing model is used to calculate the expected return from a security (ignoring unsystematic risk).

### Formula: Capital asset pricing model (CAPM)

$$R_E = R_{RF} + \beta (R_M - R_{RF})$$

#### Where:

$R_E$  = the cost of equity for a company's shares

$R_{RF}$  = the risk-free rate of return: this is the return that investors receive on risk-free investments such as government bonds

$R_M$  = the average return on market investments as a whole, excluding risk-free investments

$\beta$  = the beta factor for the company's equity shares.

The expected return from an individual security will therefore vary up or down as the return on the market as a whole goes up or down. The size of the increase or fall in the expected return will depend on:

- The size of the change in the returns from the market as a whole, and
- The beta factor of the individual security.

#### Example:

The risk-free rate of return is 4% and the return on the market portfolio is 8.5%. What is the expected return from shares in companies X and Y if:

- a) the beta factor for company X shares is 1.25
- b) the beta factor for company Y shares is 0.90?

## Alpha factor/value

The beta factor for shares is a measure of systematic risk and it ignores variations in the equity returns caused by unsystematic risk factors. When shares yield more or less than their expected return (based on the CAPM), the difference is an abnormal return. This abnormal return might be referred to as the alpha factor.

The **alpha value** can be seen as a measure of how wrong the CAPM is, it may be thought of as the difference between the expected return on a security or investment, given its systematic risk, and the actual return.

Alpha values:

- a) Reflect only temporary, abnormal returns, if CAPM is a realistic model;
- b) Can be positive or negative;
- c) Over time, will tend towards zero for any individual share, and for a well-diversified portfolio taken as a whole will be zero; and
- d) May exist due to the inaccuracies and limitations of the CAPM.

If the **alpha value is positive**, investors who do not hold shares will be tempted to buy them (to take advantage of the abnormal return), and investors who do hold shares will want to hold on to them, so share prices will rise. If the **alpha value is negative**, investors won't want to buy them, and current holders will want to sell them, so share prices will fall.

For example, ABC pic's shares have a beta value of 1.2 and an alpha value of +2%. The market return is 10% and the risk-free rate of return is 6%.

The required return is  $6\% + (10\% - 6\%) \times 1.2 = 10.8\%$

The current return = expected return  $\pm$  alpha value =  $10.8\% + 2\% = 12.8\%$



## Capital asset pricing model (CAPM) with alpha value

$$R_S = R_{RF} + \beta_S (R_M - R_{RF}) + \alpha$$

### Arbitrage Pricing Model (APM)

The CAPM is a "single factor" model. It calculates a return on an investment by relating the market risk premium to the systematic risk of the investment.

Researchers claim to have identified other factors that also affect return, including:

- company size;
- unexpected changes in interest rates;
- unexpected changes in industrial production levels; and
- unexpected inflation

Multi-factor models have been developed. The best known of these is the arbitrage pricing model. The model is similar to the CAPM in that it relates a risk premium to the underlying risk factor. However, whereas the CAPM says that there is only one such factor (systematic risk) the arbitrage pricing model says there are several. Therefore, according to this model, an investor requires a return to compensate for each of these separate risk factors.

### Formula: Arbitrage pricing model

$$R_E = R_{RF} + \beta_1 (R_1 - R_{RF}) + \beta_2 (R_2 - R_{RF}) + \beta_3 (R_3 - R_{RF}) \text{ (and so on)}$$

#### Where:

$R_E$  = the cost of equity for a company's shares

$\beta_{1,2,3}$  = the beta factor for factor 1, 2, 3 (i.e. a measure of the risk associated with the factor)

$(R_{1,2,3} - R_{RF})$  = the risk-premium associated with factor 1, 2 or 3

Systematic risk is by far the most important determinant of return, but it probably is not the only one.

### Definitions of Strategy:

There is no single definition of corporate strategy.

#### Chandler

The determination of the basic long-term goals, the objectives of an enterprise, the adoption of courses of action and the allocation of resources necessary for carrying out these goals.

#### Drucker

Strategy is a pattern of activities that seek to achieve the objectives of the organisation and adapt its scope, resources and operations to environmental changes in the long term.

#### Johnson, Scholes and Whittington

Strategy is the direction and scope of an organisation over the long term, which achieves advantage in a changing environment through its configuration of resources and competencies with the aim of fulfilling stakeholder expectations.

Each of the above definitions has a different focus but all refer to actions (courses of action, pattern of activities, configuration of resources and competencies) undertaken to achieve the objectives (aim) of an organisation.

In other words:

- a strategy consists of organised activities; and
- the purpose of these activities (the strategy) is to achieve an objective.

Strategy is long-term. Formal strategic planning by large companies, for example, might cover five years or ten years or longer into the future.

The strategic choices that an enterprise makes are strongly influenced by the environment in which the enterprise exists. The environment is continually changing, which means that strategies cannot be rigid and unchanging.

Strategies involve an enterprise in doing different things with different resources over time, as it is forced to adapt to changes in its environment.

### **Types of strategy:**

- Tactical strategy
- Operational strategy
- Financial strategy

### **Financial strategy**

Financial strategy is a component of the overall corporate strategy of an organisation.

Business strategies and action plans include financial strategies and plans.

#### **The nature of financial management**

Financial management refers to that aspect of management which involves planning and controlling the financial affairs of an organisation, to ensure that the organisation achieves its objectives (particularly its financial objectives).

This involves decisions about:

- how much finance the business needs for its day-to-day operations and for longer-term investment projects;
- where the finance should be obtained from (debt or equity) and whether it should be long term (long-term debt or equity) or short term (trade suppliers and bank overdrafts);
- how to manage short term cash surpluses and short term cash deficits;
- amounts to be paid out as dividends; and
- how to protect the organisation against financial risks.

### **Strategic financial management**

Strategic financial management refers to financial management on a bigger scale and with a longer time frame.

Strategic financial management is concerned with bigger decisions than financial management and those decisions might have a profound effect on the organisation.

Strategic financial management encompasses the full range of a company's finances including:

- setting out objectives;
- identifying resources;
- analysing data;
- making financial decisions; and
- monitoring performance to identify problems.

## **The primary corporate objective**

For companies, the primary corporate objective is often stated as:

- Wealth maximisation
- Maximising profits
- Growth in earnings per share

## **Financial objectives and corporate strategy**

Every organisation exists for a purpose. For example, an organisation might exist to manufacture electronic equipment, manufacture chemicals, provide a transport service or provide an education to children.

Within this overall purpose, an organisation should have a primary objective.

- the main objective of a company might be to maximise the wealth of the company's owners, its equity shareholders;
- the main objective of a state-owned organisation might be stated in terms of providing a certain standard of public service;
- the main objective of a charity would be to fulfil its charitable purpose (for example, provide maximum aid or support for a particular group of people).

When the main objective of an organisation is not a financial objective, there is always a financial constraint on its objective, such as providing the highest quality of public service with the available finance.

The identification of a primary objective is a starting point for the formulation of a strategy to achieve that objective.

The process can be shown as follows:

### **Strategic process**

- |         |                                                                             |
|---------|-----------------------------------------------------------------------------|
| Step 1: | Identify corporate objective (usually a financial objective)                |
| Step 2: | - Establish targets for the financial objective                             |
| Step 3: | - Develop business strategies for achieving the financial objective/targets |
| Step 4: | - Convert strategies into action plans                                      |
| Step 5: | - Monitor performance                                                       |

